

1. Show that the product of two upper triangular matrices is an upper triangular matrix.
2. For an $n \times n$ matrix $A = [a_{ij}]$, the trace is defined as $\text{tr}(A) = a_{11} + \dots + a_{nn}$. Let A and B be two $n \times n$ matrices. If $AB = \mathbf{O}$ then show that $\text{tr}((A+B)^3) = \text{tr}(A^3) + \text{tr}(B^3)$.
3. Supply two examples for each of the following and explain their geometrical meaning.
 - (a) Two linear equations in two variables with exactly one solution.
 - (b) Two linear equations in two variables with infinitely many solutions.
 - (c) Two linear equations in two variables with no solutions.
 - (d) Three linear equations in two variables with exactly one solution.
 - (e) Three linear equations in two variables with no solutions.
4. For what values of $\lambda \in \mathbb{R}$, the following system of equations has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions?

$$(5 - \lambda)x + 4y + 2z = 4, \quad 4x + (5 - \lambda)y + 2z = 4, \quad 2x + 2y + (2 - \lambda)z = 2.$$

Also, find the solutions whenever they exist.

5. Prove that the interchange of two rows of a matrix can be accomplished by a finite sequence of elementary row operations of the other two types.
6. Let A be an $n \times n$ matrix. If the system $A^2\mathbf{x} = \mathbf{0}$ has a non-trivial solution then show that the system $A\mathbf{x} = \mathbf{0}$ also has a non-trivial solution.
7. Let A and B be two 2×3 matrices that are in reduced row echelon form. If the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set then show that $A = B$.
8. Let A and B be two $m \times n$ matrices that are in reduced row echelon form. If the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set then show that $A = B$.
9. Prove or disprove: There exist two solutions $\mathbf{x}_1$ and $\mathbf{x}_2$ of some consistent non-homogeneous system $A\mathbf{x} = \mathbf{b}$ such that $\mathbf{x}_1 + \mathbf{x}_2$ is also a solution of $A\mathbf{x} = \mathbf{b}$.
10. Prove or disprove: If two matrices of the same order have the same rank then they must be row equivalent.
11. Examine whether the following sets are subspaces of $\mathbb{R}^n$ ($n \geq 3$).

$$S_1 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 + x_2 = 4x_3\}, \quad S_2 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 + x_2 \leq 4x_3\},$$

$$S_3 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 = 1 + x_2\} \quad \text{and} \quad S_4 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1x_2 = 0\}.$$

Remark: If a set S is a subspace then show that the rules

Rule 1 : $\mathbf{0} \in S$ (to claim that $S \neq \emptyset$);

Rule 2 : $\alpha\mathbf{u} + \beta\mathbf{v} \in S$ for every $\mathbf{u}, \mathbf{v} \in S$ and $\alpha, \beta \in \mathbb{R}$.

are satisfied. If S is **not** a subspace then show that either $\mathbf{0} \notin S$ or show, *by a counter-example*, that **Rule 2** is not satisfied.

12. Find all the subspaces of $\mathbb{R}^2$.
13. Show that $W = \{[x, y, z, w]^t \in \mathbb{R}^4 : w - z = y - x\}$ is a subspace of $\mathbb{R}^4$, spanned by the vectors $[1, 0, 0, -1]^t$, $[0, 1, 0, 1]^t$ and $[0, 0, 1, 1]^t$. Also, determine $\dim(W)$.
14. Find three vectors in $\mathbb{R}^3$ which are linearly dependent but any two of them are linearly independent.
15. Let $S = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$, where $\mathbf{x}_1 = [1, 0, 0, 0]^t$, $\mathbf{x}_2 = [1, 1, 0, 0]^t$, $\mathbf{x}_3 = [1, 2, 0, 0]^t$ and $\mathbf{x}_4 = [1, 1, 1, 0]^t$. Determine all $\mathbf{x}_i \in S$ such that $\text{span}(S) = \text{span}(S \setminus \{\mathbf{x}_i\})$.
16. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be linearly dependent vectors in $\mathbb{R}^n$ and let $\mathbf{v}, \mathbf{w}, \mathbf{x}$ be linearly independent vectors in $\mathbb{R}^n$, where $n \geq 3$. Show that (i) $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$, and (ii) $\mathbf{x}$ is not a linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$.
17. Let $M = \{[x, y, z]^t \in \mathbb{R}^3 : x + y + 4z = 0\}$ and $N = \{[x, y, z]^t \in \mathbb{R}^3 : x + y + z = 0\}$. Show that M and N are subspaces of $\mathbb{R}^3$. Determine a basis for each of $M, N, M + N$ and $M \cap N$. Also, interpret your result geometrically.
18. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set of a vector space V . Write $\mathbf{w}_1 = \mathbf{v}_1$ and $\mathbf{w}_k = \sum_{i=1}^k c_{ik} \mathbf{v}_i$ for $2 \leq k \leq n$ such that $c_{ik} > 0$ for all $1 \leq i \leq n, 2 \leq k \leq n$. Show that the set $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ is also linearly independent.
19. Consider the vector space $\mathbb{R}_4[x]$ of all polynomials with real coefficients of degree at most 4. Show that $\{p(x) \in \mathbb{R}_4[x] : p(1) = p(-1) = 0\}$ is a subspace of $\mathbb{R}_4[x]$. Determine a basis and the dimension of this subspace.
20. Let $M_n(\mathbb{R})$ denote the vector space of all $n \times n$ real matrices. Show that the set $S_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : A = A^t\}$ is a subspace of $M_n(\mathbb{R})$. Also, find a basis and the dimension of $S_n(\mathbb{R})$.
21. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a basis for a vector space V . Show that the set $\{\mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_2 + \mathbf{x}_3, \dots, \mathbf{x}_n + \mathbf{x}_1\}$ is also a basis for V if and only if n is an odd integer.
22. Let W_1 and W_2 be two subspaces of a finite dimensional vector space V . Show that $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$.
23. Let V be a finite dimensional vector space and let V_1 and V_2 be two subspaces of V . If $\dim(V_1 + V_2) = \dim(V_1 \cap V_2) + 1$, show that either $V_1 + V_2 = V_1, V_1 \cap V_2 = V_2$ or $V_1 + V_2 = V_2, V_1 \cap V_2 = V_1$. [Equivalently, for subspaces V_1 and V_2 of V , if neither contains the other, then $\dim(V_1 + V_2) \geq \dim(V_1 \cap V_2) + 2$].
24. Find the coordinates of the vector $[1, 2, 3]^t$ with respect to the bases B and C for $\mathbb{R}^3$, where

$$B = \{[1, 1, 0]^t, [0, 1, 1]^t, [1, 0, 1]^t\} \text{ and } C = \{[1, 1, 1]^t, [1, 1, -1]^t, [1, -1, 1]^t\}.$$

Also, find the matrix P such that $[\mathbf{x}]_B = P[\mathbf{x}]_C$ for all $\mathbf{x} \in \mathbb{R}^3$.

25. Let A be an $n \times m$ matrix and let B be an $m \times n$ matrix. Prove that the matrix $I_m - BA$ is invertible if and only if the matrix $I_n - AB$ is invertible.

26. Let A be a nilpotent matrix (*i.e.*, $A^m = \mathbf{O}$ for some $m \geq 1$). Show that the matrix $I + A$ is invertible.
27. Let A and B be two $n \times n$ matrices. If AB is invertible then show that each of the matrices A and B must be invertible, in at least three different ways.
28. Let A be an $n \times n$ skew-symmetric matrix. Prove that $\mathbf{x}^t A \mathbf{x} = 0$ for all $\mathbf{x} \in \mathbb{R}^n$. Further, prove that the matrix $I_n + A$ is invertible.
29. Check, without calculating the actual value, whether the determinant of the following matrix is even or odd:

$$\begin{bmatrix} 2 & 2 & 5 & 4 \\ 5 & 3 & 2 & 2 \\ 2 & 5 & 7 & 5 \\ 1 & 3 & 9 & 2 \end{bmatrix}.$$

30. Find the determinant of the following matrices:

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 2 & 2 & 3 & 4 & \dots & n \\ 3 & 3 & 3 & 4 & \dots & n \\ \dots & \dots & \dots & \dots & \dots & \dots \\ n & n & n & n & \dots & n \end{bmatrix}.$$

31. Let $A = [a_{ij}]$ be an $n \times n$ matrix such that $\sum_{j=1}^n a_{ij} = 1$ for all $i = 1, 2, \dots, n$. Is it true that $\det(A - I_n) = 0$? Justify your answer.
32. Let $A = [a_{ij}]$ be an invertible matrix and let $B = [k^{i-j} a_{ij}]$, where $k (\neq 0) \in \mathbb{R}$. Find B^{-1} and $\det(B)$.
33. For all invertible matrices A and B of the same size, show that $\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$. (The result is also true for non-invertible matrices, but the proof is beyond the present scope of this course.)
34. If A is an $n \times n$ matrix then prove that $\det(\text{adj}(A)) = (\det A)^{n-1}$.
35. Let A and B be two matrices, where $\text{rank}(A) = r$. Show that
- if AB is defined then $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$;
 - there exist invertible matrices T, S such that

$$TAS = \begin{bmatrix} I_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix};$$

- there exist matrices $P_{m \times r}, Q_{r \times n}$ such that $A = PQ$; (this is known as Rank-Factorization Theorem)
- A can be expressed as a sum of r rank one matrices;
- $r = 1$ if and only if $A = \mathbf{u}\mathbf{v}^t$ for some $\mathbf{u} (\neq \mathbf{0}) \in \mathbb{R}^m$ and $\mathbf{v} (\neq \mathbf{0}) \in \mathbb{R}^n$;

(f) if $A + B$ is defined then $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

36. Let $A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 6 & 4 & 8 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$. Determine all $\mathbf{b} \in \mathbb{R}^3$ for which the system $A\mathbf{x} = \mathbf{b}$ is consistent.

37. Examine whether the following maps $T : V \rightarrow W$ are linear transformations.

(a) $V = W = \mathbb{C}^2(\mathbb{C})$ and $T[z_1, z_2]^t = [\bar{z}_1, \bar{z}_2]^t$ for all $[z_1, z_2]^t \in \mathbb{C}^2$.

(b) $V = W = M_n(\mathbb{R})$ and fix $B \in M_n(\mathbb{R})$. Consider $T(A) = AB - BA$ for all $A \in V$.

38. Let $T : V \rightarrow V$ be a linear transformation such that $T \circ T = I$ and let $\mathbf{v} \in V$.

(a) Show that $\{\mathbf{v}, T(\mathbf{v})\}$ is linearly dependent if and only if $T(\mathbf{v}) = \pm\mathbf{v}$.

(b) Give an example of such a linear transformation with $V = \mathbb{R}^2$.

39. Find bases for the range space and the null space of the linear transformation $T : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ defined by $T(A) = AB - BA$ for all $A \in M_2(\mathbb{R})$, where $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Hence find the rank and the nullity of T .

40. Let V and W be two finite-dimensional vector spaces and let $T : V \rightarrow W$ be a linear transformation. Show that

(a) if $\dim(V) < \dim(W)$ then T is not onto; and

(b) if $\dim(V) > \dim(W)$ then T is not one-one.

41. Let T be a linear transformation on a vector space V and let $\dim V = n$. If

$$T^{n-1}(\mathbf{x}) \neq \mathbf{0} \text{ but } T^n(\mathbf{x}) = \mathbf{0} \text{ for some } \mathbf{x} \in V,$$

then show that the set $\{\mathbf{x}, T(\mathbf{x}), \dots, T^{n-1}(\mathbf{x})\}$ is a basis for V . Also, find the matrix representation of T with respect to this basis.

42. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, defined by

$$T[x, y, z]^t = [2x + 3y - z, 4x - y + 2z]^t \text{ for all } [x, y, z]^t \in \mathbb{R}^3.$$

Find $[T]_{C \leftarrow B}$, where $B = \{[1, 1, 0]^t, [1, 2, 3]^t, [1, 3, 5]^t\}$ and $C = \{[1, 2]^t, [2, 3]^t\}$.

43. Let A be an $n \times n$ matrix. Find the eigenvalues of $A - 3I$ in terms of the eigenvalues of A . Also, show that their corresponding eigenspaces are equal.

44. Let $A = [a_{ij}]$ be an $n \times n$ matrix and let $k \in \mathbb{R}$. Suppose that $\sum_{j=1}^n a_{ij} = k$ for $i = 1, 2, \dots, n$. Show that k is an eigenvalue of A . Find an eigenvector of A corresponding to the eigenvalue k .

45. Find all real values of a, b, c, d, e, f for which the matrix $\begin{bmatrix} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is diagonalizable.

46. Let A be a 6×6 matrix with characteristic polynomial $p(\lambda) = (1 + \lambda)(1 - \lambda)^2(2 - \lambda)^3$.

- (a) Prove that it is not possible to find three linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^6$ such that $A\mathbf{v}_1 = \mathbf{v}_1, A\mathbf{v}_2 = \mathbf{v}_2$ and $A\mathbf{v}_3 = \mathbf{v}_3$.
- (b) If A is diagonalizable, find the dimensions of the eigenspaces E_{-1}, E_1 and E_2 ?
47. Let A and B be two $n \times n$ matrices satisfying $AB = BA$ and let B have n distinct eigenvalues. Show that the matrix A is diagonalizable.
48. Let $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$. Prove that
- (a) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ (**Cauchy-Schwarz inequality**);
- (b) $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent;
- (c) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.
49. Let $V = \{[x, y, z, w]^t \in \mathbb{C}^4 : x = z + iw, y = z - w\}$. Show that V is a subspace of $\mathbb{C}^4$. Find a basis for each of V and $V^\perp$.
50. Let A be a 2×2 orthogonal matrix. Show that there exists a real number θ such that $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ or $A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$. In the first case, A rotates the vectors of $\mathbb{R}^2$ by the angle θ counterclockwise, and in the second case, A reflects the vectors of $\mathbb{R}^2$ about a line; in this case find the line.
51. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthonormal basis of $\mathbb{C}^n$. Show that for any $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$,
- $$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n (\mathbf{x} \cdot \mathbf{v}_i)(\mathbf{v}_i \cdot \mathbf{y}) \quad \text{and} \quad \mathbf{x} \cdot \mathbf{x} \geq \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2 \quad \text{for } 1 \leq k \leq n.$$
- Further, show that $\mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2$ iff $\mathbf{x} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$, where $1 \leq k \leq n$.
52. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let $A = [a_{ij}]$ be the matrix of T with respect to an orthonormal basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of $\mathbb{R}^n$. Show that $a_{ij} = \mathbf{v}_i \cdot (T\mathbf{v}_j)$ for all i, j .
53. Let W be a subspace of $\mathbb{C}^n$ and let $\mathbf{u} \in \mathbb{C}^n$. Show that $\mathbf{v} \in W$ is the projection of $\mathbf{u}$ onto W if and only if $\|\mathbf{u} - \mathbf{v}\| \leq \|\mathbf{u} - \mathbf{w}\|$ for every $\mathbf{w} \in W$.
54. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal set of non-zero vectors in $\mathbb{C}^n$. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q$ be some vectors in $\mathbb{C}^n$ that are orthogonal to S . If $p + q > n$ then show that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q$ are linearly dependent.
55. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n, n \geq 2$ such that $\|\mathbf{v}\| = \|\mathbf{w}\| = 1$. Prove that there exists an orthogonal matrix A such that $A(\mathbf{v}) = \mathbf{w}$ and $\det(A) = 1$.
56. (Spectral Decomposition Theorem) Let A be a normal matrix. Show that there are distinct complex numbers $\alpha_1, \dots, \alpha_k$ and matrices $E_1, \dots, E_k$ such that $A = \alpha_1 E_1 + \dots + \alpha_k E_k$, where $E_1 + \dots + E_k = I$, $E_i E_j = O$ for $i \neq j$ and $E_i^2 = E_i$ for $i = 1, \dots, k$.
57. Orthogonally diagonalize, and then find spectral decomposition of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

58. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of a matrix A . Show that $\det(A) = \prod_{i=1}^n \lambda_i$ and $\text{trace}(A) = \sum_{i=1}^n \lambda_i$.
59. Find a 3×3 symmetric matrix with eigenvalues 1, 2 and 3, and corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.
60. Let $\mathbf{q}$ be a unit vector in $\mathbb{R}^n$ and $W = \text{span}(\mathbf{q})$. Show that the orthogonal projection of a vector $\mathbf{v}$ onto W is given by $(\mathbf{q}\mathbf{q}^t)\mathbf{v}$, and that the matrix of this projection map is $\mathbf{q}\mathbf{q}^t$.
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